

Manish Choudhary

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Location: Hoshiarpur, Punjab, India

EDUCATION

- Guru Nanak Dev Engineering College** Ludhiana, Punjab, India
• *Bachelor of Technology – Computer Science & Engineering; CGPA: 6.89/10* 2022 – 2026

EXPERIENCE

- Data Science Intern** Sabudh Foundation, Mohali, Punjab
• *Gemini API, MySQL, Streamlit, Serper API, ReportLab, Python* Jul 2025 – Jan 2026
 - Led a 5-member team to build an automated student assessment system for Sabudh Foundation, replacing a manual spreadsheet workflow (Google Forms/KoboToolbox); architected an async **Gemini API** pipeline that evaluates multi-domain competency scores and serializes narrative reports into a normalized **7-table MySQL schema** with a Check-or-Create cache (<0.5s retrieval) to prevent redundant LLM calls.
 - Engineered a context-aware Q&A chatbot using **multi-table SQL JOINS** to fetch student records, integrated **Serper API** (web search RAG) to dynamically recommend verified course links for identified skill gaps without LLM hallucinations, and added 22 Indian language translations (**langdetect** + **deep-translator**) with **RBAC** authentication.
 - Shipped an interactive **Streamlit** web platform with automated **ReportLab PDF report cards** for **100+ NGO students**, establishing the organization's first data-driven evaluation pipeline and eliminating ~4–5 hours/week of manual volunteer scoring.

PROJECTS

- AI Music Generation Platform** github.com/manish-01882/ai-music-gen
Python, FastAPI, Modal (Serverless GPUs), PyTorch, HuggingFace Diffusers/Transformers, Next.js 15, AWS S3, Inngest, Prisma, Neon Postgres
 - Orchestrated 3 AI models (**ACE-Step**, **Qwen2-7B**, **SDXL-Turbo**) on **Modal serverless GPUs** with **bfloat16/fp16** precision, container warmup, and persistent volume caching to eliminate cold-start downloads.
 - Built a 2-stage **LLM preprocessing pipeline** (Qwen2-7B) converting free-text prompts into structured audio tags and **[verse]/[chorus]/[bridge]** lyrics for 3-minute music generation.
 - Designed an async task pipeline with **Inngest** (concurrency control, auto-retry) and a **Prisma/Neon Postgres** state machine tracking 30–60s GPU jobs through **queued/processing/completed** lifecycle.
 - Delivered cloud media via **AWS S3** presigned URLs and a **Zustand** global audio player persisting playback state across Next.js routes.
- TB-MTNet: AI-Powered Tuberculosis Detection from Chest X-Rays** github.com/manish-01882/tb-mtnet
PyTorch, Inception-v3, Transformer Encoder, U-Net, FastAPI, Grad-CAM
 - Built an end-to-end medical imaging system that automatically analyses chest X-rays to detect Tuberculosis and estimate severity – simultaneously classifying TB presence and predicting a severity score using a dual-head model (Inception-v3 + Transformer encoder).
 - Trained on ~12,000 chest X-rays; achieved **0.983 AUROC** for TB detection and **8.51 MAE** for severity regression, with a custom 3-tier labeling pipeline to resolve inconsistent annotations across the dataset.
 - Deployed as a **FastAPI** service with **U-Net lung segmentation** and **Grad-CAM heatmap overlays** so clinicians can see exactly which region of the X-ray the model is reacting to.

SKILLS

- Languages:** Python, C/C++, SQL, JavaScript/TypeScript
- Data & Databases:** Pandas, NumPy, MySQL/Postgres, MongoDB
- Libraries & Frameworks:** Deep Learning Frameworks: (TensorFlow, Keras, PyTorch), **Classical ML (Scikit-Learn, Scipy, NLTK)**, Neural Architectures (**CNNs, Transformers, LSTMs, RNNs**), Generative AI & LLMs
- Web / API:** FastAPI, Next.js, React, Streamlit
- Tools:** Git, Docker

RELEVANT COURSEWORK

- Courses:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, Amazon Web Services (AWS), Statistics, Linear Algebra

CERTIFICATIONS

- Data Science Training** – Sabudh Foundation